American Computer Science League

2023-2024 ● Contest 1: Short Problems ● Intermediate Division

1. Computer Number Systems

A palindrome is a word, phrase, or number that is the same forwards and backwards. For example, *madam* and *12321* are palindromes.

What is the next year in our calendar when its octal representation is a palindrome?

A. 2032
B. 3773
C. 4004
D. 2043
E. 2112

2. Computer Number Systems

Convert the following hex number to octal. What octal digit occurs most often?

$F43A6C3091_{16}$

A. 0
B. 1
C. 2
D. 3
E. 7

3. Recursive Functions

Find $f(17, 11)$ given:

$$f(x,y) = \begin{cases} f(x-1, y-2) + 3 & \text{if } x < y \\ f(y-1, x-2) + 1 & \text{if } x > y \\ x - y & \text{if } x = y \end{cases}$$

A. 8
B. 9
C. 12
D. 15
E. 16

4. Recursive Functions

Find $f(24)$ given:

$$f(x) = \begin{cases} 2 \cdot f(x-4) & \text{if } x \geq 18 \\ f(x+1) - 1 & \text{if } 15 < x < 18 \\ x + 2 & \text{if } x \leq 15 \end{cases}$$

A. 16
B. 32
C. 60
D. 120
E. 240

5. What Does This Program Do? (Branching)

Sales employees are often paid based on commission. Using the following conditions:

sales < 5000, 1% of sales
5000 ≤ sales < 10000, 1% of the first $5,000 and 3% of the rest
sales ≥ 10000, 1% of the first $5,000, 3% of the next $5,000, and 5% of the amount over $10,000

List all the lines of code (a,b,c,d) that will correctly calculate a salesperson's commission (stored in variable `cpay`) for any positive value of sales (given in variable `sales`).

a.
```
if sales >= 10000 then
    cpay = sales*.05 + 5000*.03 + 5000*.01
else if sales >= 5000 then
    cpay = sales*.03 + 5000*.01
else
    cpay = sales*.01
end if
```

b.
```
if sales < 5000  then
    cpay = sales*.01
else if sales < 10000 then
    cpay = (sales-5000)*.03 + 5000*.01
else
    cpay = (sales-10000)*.05 + 5000*.03 + 50
end if
```

c.
```
if sales < 5000  then
    cpay = sales*.01
end if
if sales >= 5000 && sales < 10000 then
    cpay = (sales-5000)*.03 + 5000*.01
else if sales >= 10000 then
    cpay = (sales-10000)*.05 + 5000*.03 + 50
end if
```

d.
```
if sales >= 10000 then
    cpay = (sales-10000)*.05 + 5000*.03 + 50
else if sales >= 5000 then
    cpay = (sales-5000)*.03 + 5000*.01
else
    cpay = sales*.01
end if
```

A. b, c, d
B. a, c, d
C. a, b, c
D. a, b, d
E. a, b, c, d

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1. 计算机数字系统

任何从前往后或从后往前读都相同的单词、短语、数字都为回文。例如，*madam* 和*12321* 都是回文。

请问日历中接下来哪一年的八进制表达为回文？

A. 2032
B. 3773
C. 4004
D. 2043
E. 2112

2. 计算机数字系统

将下列的十六进制数转换为八进制。在八进制中哪个数字出现的频率最高？

$F43A6C3091_{16}$

A. 0
B. 1
C. 2
D. 3
E. 7

3. 递归函数

根据下列表达式，求 $f(17, 11)$ ：

$$f(x,y) = \begin{cases} f(x-1,y-2)+3 & \text{如果 } x < y \\ f(y-1,x-2)+1 & \text{如果 } x > y \\ x - y & \text{如果 } x = y \end{cases}$$

A. 8
B. 9
C. 12
D. 15
E. 16

4. 递归函数

根据下列表达式，求 $f(24)$

$$f(x) = \begin{cases} 2 \cdot f(x-4) & \text{如果 } x \geq 18 \\ f(x+1) - 1 & \text{如果 } 15 < x < 18 \\ x + 2 & \text{如果 } x \leq 15 \end{cases}$$

A. 16
B. 32
C. 60
D. 120
E. 240

5. 这个程序是做什么的？（分支）

销售人员的工资通常是基于其所赚佣金而定。情况如下：

销售额 < 5000 时, 佣 金 = 销售额的 1% ；
5000 ≤销售额 < 10000 时, 佣 金 = 前 $5,000 的 1% + 剩下部分的 3%；
销售额 ≥ 10000时, 佣 金 = 第一个 $5,000 的 1% + 第二个 $5,000 的 3% + 超过$10,000的部分的5% 。

请给出以下代码块中能由任何正值销售额（由变量sales给定）正确计算出销售人员佣金（储存在变量 cpay 中）的代码块。

a.
```
if sales >= 10000 then
    cpay = sales*.05 + 5000*.03 + 5000*.01
else if sales >= 5000 then
    cpay = sales*.03 + 5000*.01
else
    cpay = sales*.01
end if
```

b.
```
if sales < 5000  then
    cpay = sales*.01
else if sales < 10000 then
    cpay = (sales-5000)*.03 + 5000*.01
else
    cpay = (sales-10000)*.05 + 5000*.03 + 50
end if
```

c.
```
if sales < 5000  then
    cpay = sales*.01
end if
if sales >= 5000 && sales < 10000 then
    cpay = (sales-5000)*.03 + 5000*.01
else if sales >= 10000 then
    cpay = (sales-10000)*.05 + 5000*.03 + 50
end if
```

d.
```
if sales >= 10000 then
    cpay = (sales-10000)*.05 + 5000*.03 + 50
else if sales >= 5000 then
    cpay = (sales-5000)*.03 + 5000*.01
else
    cpay = sales*.01
end if
```

A. b, c, d
B. a, c, d
C. a, b, c
D. a, b, d
E. a, b, c, d